Measuring the roughness exponent of spin coated PMMA thin films

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Polymer thin films and spin coat polymer films are essential to current research in organic semiconductors as cost-effective and flexible replacements for inorganic components. In this experiment, PMMA thin films were produced using spin coating, and the roughness exponent of the films were measured using an atomic force microscope (AFM). The height data of the resulting topographical images were extracted, and a roughness exponent of $\alpha = 0.65 \pm 0.02$ was calculated at system sizes of 1 μm $\times$ 1 μm. This value is close to the 2 dimensional roughness exponent for the universality class NCN4. Additionally, experimental results for system sizes of 10 μm $\times$ 10μm found that saturation did not occur as expected, clearly exhibiting complex intersections of fluid dynamics and surface growth processes.

I. INTRODUCTION

The morphology of polymer thin films is an important and relevant topic with significant ramifications on the future of organic semiconductor and microelectronic devices[1]. An understanding of morphological parameters and their effects on thin film quality can lead to improvements including upgraded adhesive properties, enhanced conductivity, higher reliability, and more mechanical durability in polymer thin films. Spin coating, a popular method by which to produce thin films, has applications in OLED, fuel cell, and optical coating technologies[2]. Understanding what factors influence the thin film morphology requires knowledge of universality classes.

When considering a surface, there are many indistinguishable factors that can influence growth. Universality classes attempt to simplify these to models which obey comprehensible, limited laws for given growth processes. Universality classes are characterized by symmetry and conservation laws which govern their growth equations. The roughness of an interface is expressed by the interface width,

$$w(L,t) = \sqrt{\frac{1}{L}\sum_{i}^{L}\left[h(i,t) - \bar{h}(t)\right]^2}. \tag{1}$$

Here, L is the total system size, and t is the time at which the interface is examined, given it's growth as a function of time. h and $\bar{h}$ refer to the height at a given point and the mean height, respectively.

Eq.(1) describes the process by which a surface grows and roughens as the material is deposited on the surface. This process is split into two regimes: the growth regime and the saturation regime. The saturation regime of the interface width w_{sat} is dependent on the size of the system. As L increases, w_{sat} also increases, following the power law:

$$w_{\text{sat}}(L) \propto L^{\alpha}, \tag{2}$$

where exponent α is named the roughness exponent. α characterizes the roughness of a saturated surface given the size of the system. The roughness of an interface is expected to grow for some time before saturating. The time at which a surface is saturated is described by the crossover time or saturation time t_x,

$$t_x \propto L^z. \tag{3}$$

Here, z represents the dynamical exponent. The crossover time describes the switching of behavior from the growth regime to the saturation regime. The growth regime is governed by the growth exponent β and describes how the roughening process occurs as a time dependent dynamic. The growth regime is described by the power law:

$$w(L,t) \propto t^{\beta}. \tag{4}$$

The dynamical exponent z relates α to β, such that $z = \alpha/\beta$, insinuating that the two regimes are not independent. In our experiment, we aim to extract α by utilizing the height-height correlation function (HHCF):

$$g(r) = \langle [h(x) - h(x-r)]^2 \rangle. \tag{5}$$

Here, x is the position of the height in question, and r is the displacement vector[3]. Eq. (1) characterizes the roughness saturation across the entire interface, but we can show that saturation as a function of separation using Eq. (5). In the small length scale limit however, we can approximate Eq. (5) as

$$g(R) \approx AR^{2\alpha}, \tag{6}$$

where A is some constant and R is a 2 dimensional length scale, characterized as

$$R = \sqrt{L_x^2 + L_y^2}.$$

The 3 critical exponents form the core of a universality class and characterize the behavior of a surface. Known universality classes arise from the determination of values for α, β, and z, and interfaces which obey the same conservation and symmetry laws, and therefore have the same critical exponents, will belong to the same universality class. In this experiment, the focus is to determine the value of α for spin coated PMMA thin films as one step towards determining their universality class. After the process of spin coating PMMA films, the films were imaged using an atomic force microscope to gather direct data on the nano-topography of the film surfaces. The height-height correlation of points on the topography were calculated as a function of the length scale of the image, from which α could be derived.

In this paper, we use the method of spin coating to produce thin PMMA films. We find the height data of our films using the AFM, from which we are able to perform a height-height correlation of points on the film. We can take a linear fit of this data in order to derive the roughness exponent that characterizes the surface of our films.

II. EXPERIMENTAL METHODS

Natural oxide silicon wafer substrates were first placed in a UV ozone oven to prepare the substrate. This process was followed by submersion in an ultrasonic bath of acetone for a 10 minute duration. This process was repeated with a 70% solution of isopropyl (C_3H_8O) and water. This process was used to minimize any debris that would have collected on the wafers surface and affect the thin film surface[4].

In preparation for spin coating, polymethyl methacrylate (PMMA) was chosen as the polymer due to it's wide availability and the generalized properties it possesses. PMMA was dissolved in toluene ($C_6H_5CH_3$) at a 1:10 PMMA/toluene ratio by mass. The solution was mixed on a hotplate at 50°C for two hours. After this, the solution was left to mix overnight at room temperature, at which point the solution was thoroughly homogenized and clear. The PMMA-toluene solution was then dropped onto a cleaned silicon wafer and placed in the spin coater. The wafers were spin coated for 1 minute at 2000 rpm. Following the spin coating, the wafers were left to let remaining toluene evaporate. The evaporation will introduce some uncertainty by creating random noise on the surface of our film.

The surface morphology of the films was measured using an Asylum Research model MED-3D atomic force microscope (AFM), pictured in Fig. 1, in AC Air Topography mode. Square topographical images were taken at 1, 5, and 10 μm length by width scales, which we refer to as length scales. This variation in size was performed to fully capture the entirety of the saturation regime. Images were taken over several regions and different films in order to ensure no anomalous regions were observed. Images were captured at a 512×512 pixel resolution. Imaging was performed on silicon wafers before and after the cleaning process to ensure that not only would the cleaning not denature the surface of the wafer, but also the surface of the films grown on top.

Resulting AFM images were analyzed using Eq. (5), and a log-log plot of the HHCF produced a graph that modeled how the roughness of the film saturated as a function of R. The pre-saturation points were linearly modeled using a function fitting program, with the slope of the fit equal to 2α. To perform the linear fit, we used the Numpy object Polyfit in a Python environment.

FIG. 1. Asylum Research model MED-3D AFM in the North Carolina State University Education and Research Lab.

III. RESULTS

Images produced by the AFM show a heat map of height, seen in Fig. 2. A text file containing height data was extracted from each of the 512×512 pixels. The height data was used with the HHCF to show saturation as a function of the 2 dimensional length scale R, as shown in Fig. 3. The linear fit slope of the pre-saturation points, seen in Fig. 4, was used to find the roughness exponent. Polyfit returns the variance of each fit, from which we derived the error of the fit. The calculated roughness exponents and error of each image are displayed in Table I.

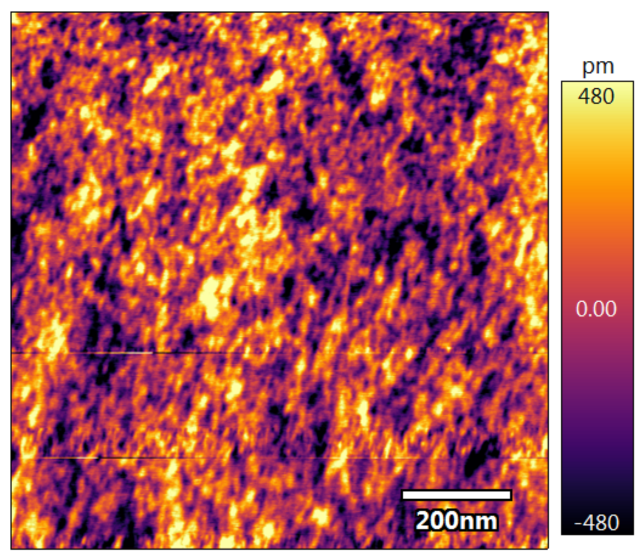

FIG. 2. 1 μm image of film 1 taken from AFM.

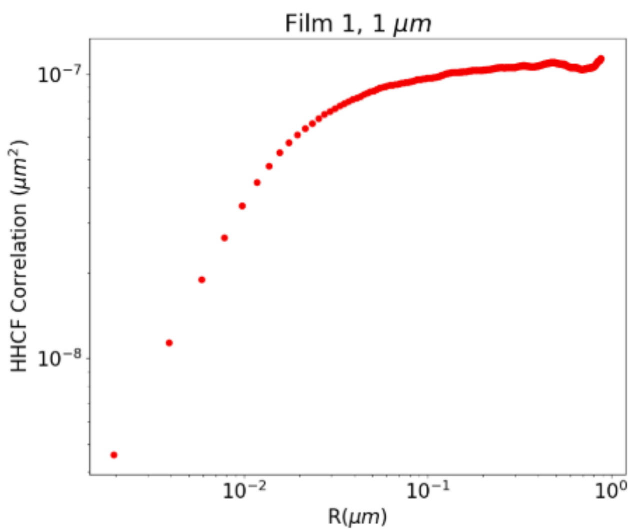

FIG. 3. A log-log plot of the results of the HHCF on the 1 μm scale height data for film 1.

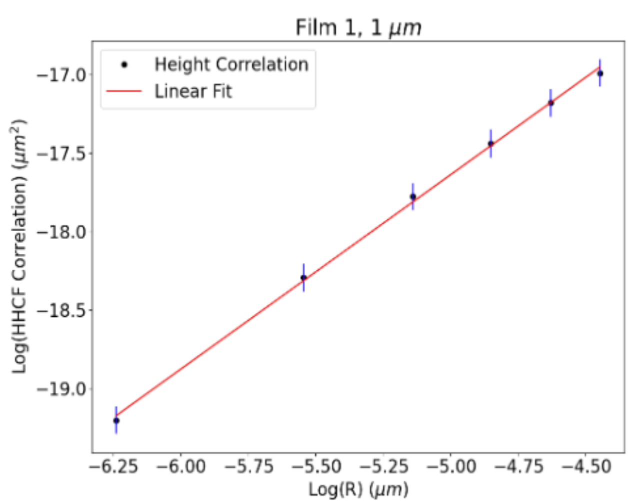

FIG. 4. A linear fit of the pre-saturation points for the film 1, 1 μm scale HHCF results. The error bars in this graph are scaled 10x.

TABLE I. Calculated slopes and errors across all length scales for both films.

	Size (μm x μm)	α	Error ($\pm$)
	1	0.62	0.01
	5	0.49	0.05
Film 1	10	0.33	0.03
	1	0.67	0.02
	5	0.33	0.03
Film 2	10	0.55	0.02

IV. DISCUSSION

Upon analysis of the HHCF graphs across all length scales, interesting behavior emerges at the 10 μm scales. Although we observed that the 1 and 5 μm length scales showed saturation as R increases, we observed that the 10 μm scales shows no such saturation, as seen in Fig 5 and Fig 6. We hypothesize that the behavior at these large length scales originates from ...

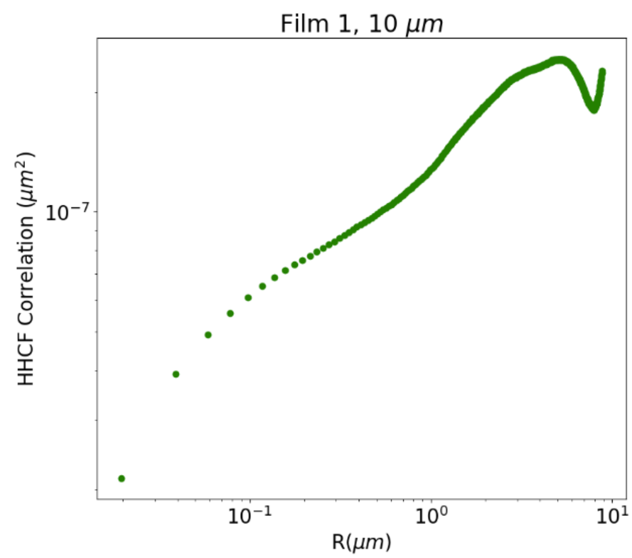

FIG. 5. HHCF results for 10 μm lengths scale of film 1, displaying no saturation with R.

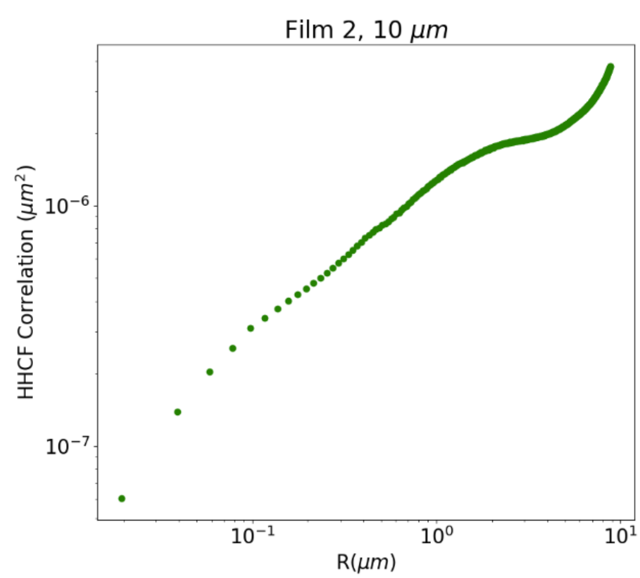

FIG. 6. HHCF results for 10 μm lengths scale of film 2, displaying no saturation with R.

In an effort to minimize these effects, we have decided to prioritize data on the 1 μm length scale, for which we calculated that $\alpha = .65 \pm .02$. Initially we had expected that we would find α equal to 1, which would suggest it is part of the LCN4 universality class, meaning linear conservative dynamics and non-conservative noise[1]. Instead these results are much closer to the NCN4 universality class where $\alpha = 2/3$ for a 2 dimensional interface. NCN4 universality class represents materials governed by non-conservative linear dynamics and non-conservative noise. If this were true, then we could expect to find that β would be 1/5 and $z = 10/3$. Interestingly, this would coincide with the critical exponents of another method of polymer deposition, vapor deposition[5]. This would suggest that despite being 2 different methods of film growth, the long-term behavior and roughness of spin coating and vapor deposition could be statistically very similarly.

The error mentioned in our final calculation of α is calculated from the covariance matrix produced by the Polyfit object. This error represents error in the data analysis method, for which there is little. Other sources of statistical error arise from background noise affecting the AFM readings and evaporative noise produced by the evaporative effects of the toluene. This noise matches speculation of our universality class as evaporative noise is non-conservative. Systematic error might occur in the process of imaging the films or throughout the process of spin-coating.

Examination of α and potential sources of error is influenced by the unexpected tendency of larger system sizes (> 10 μm) to not reach a point of saturation. Upon examination of Fig. 5 and Fig. 6, it is apparent that a system of length scale 10 μm does not saturate and appears to linearly increase in roughness as system size increases. This behavior is abnormal as theory dictates that with saturation occurring at smaller system sizes (1 μm and 5 μm) that similar saturation should occur at larger length scales. We believe that this abnormality is a symptom of larger forces at play rather than error in the methodology. The cause of the non-saturation may occur from the intersection of surface growth concepts and fluid mechanics. During film growth, standing waves or other disturbances may disrupt film growth formation at large length scales - leading to the appearance of non-saturation. A solution to this complication is not apparent as the definite origin has not been determined.

As part of our examination, we find that values of α vary over system size. This is an unexpected trend as our theoretical background tells us that α should remain consistent across system size. Changes in α can represent changes in the roughness of an interface[6]. Rather than believe that this trend represents an intrinsic property of the material, it is more likely that it results from decreasing image resolution at larger length scales. As a constant pixel count is chosen, larger system sizes have worse resolution than smaller systems. At 1 μm, a single pixel represents 3.815 nm^2 while a pixel at 10 μm is 381.9 nm^2. Worsening pixel resolution causes less available points to be available pre-saturation. This effect is exacerbated by system size negatively affecting this quality as well. Resulting linear fits done at 10 μm have very points, leading to a worse quality fit. Due to the low number of available data points, post-saturation points may be used and may cause lower α values than expected. To solve this problem, images of higher resolution must be used.

Our next steps would include exploring how factors such as polymer to solvent ratio, spin coating parameters, and methods of setting the film affect surface growth. We know that these conditions can affect the growth and thickness of the film[7], and determining an effective way to model the growth of the film as a function of time is very important for calculating the growth exponent β. Using β, one could also calculate the dynamic exponent z and use all three critical exponents to fully characterize the universality class of spin coated PMMA thin films. In addition to this, we feel that it would be beneficial to repeat height measurements at a 1024×1024 pixel resolution instead, as this would give us many more pre-saturation points to linear fit.

V. SUMMARY AND CONCLUSION

PMMA thin films were produced on natural oxide silicon wafer substrates using the process of spin coating. These films were analyzed using atomic force microscopy to extract the height data of the surface of the films across 3 different length scales. We then used a height-height correlation function to plot the roughness saturation as a function of distance between points on the surface of the films. Finally, we took a linear fit of pre-saturation points in order to calculate the roughness exponent of our film surfaces.

We calculated that for 1 μm $\times$ 1 μm length scales, the roughness exponent alpha was 0.65 $\pm$ .02, which could be consistent with the universality class NCN4 although we cannot say for sure until the growth exponent β and the dynamic exponent z are explored. Convergence with NCN4 however would imply that there are statistical similarities between the methods of spin coating and vapor deposition in film growth. Interestingly too, we also found that at the 10 μm $\times$ 10 μm length scales, the surface roughness never saturated as a function of distance between points, which is inconsistent with our model. We believe this arises from standing waves on the surface of the film that appear during the spin coating process, and only manifest in the data at the larger length scales.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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